

Induction and recursion

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Chapter Summary

- Mathematical Induction
- Strong Induction
- Well-Ordering
- Recursive Definitions

Section Summary

- Mathematical Induction
- Examples of Proof by Mathematical Induction
- Mistaken Proofs by Mathematical Induction
- Guidelines for Proofs by Mathematical Induction

Climbing an infinite ladder

Motivation:

Suppose that we have an infinite ladder:

- 1 We can reach the first rung of the ladder.
- 2 If we can reach a particular rung of the ladder, then we can reach the next rung.

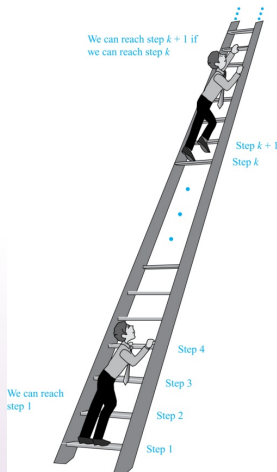
From (1), we can reach the first rung.

Applying (2), we can reach the second rung.

Applying (2) again, we can reach the third rung.

And so on.

We can apply (2) any number of times to reach any particular rung, no matter how high up.



Principle of Mathematical Induction

Principle of Mathematical Induction:

To prove that $P(n)$ is true for all positive integers n :

Basis Step: Show that $P(1)$ is true.

Inductive Step: Show that $P(k) \rightarrow P(k+1)$ is true for all positive integers k .

To complete the inductive step, assuming the inductive hypothesis that $P(k)$ holds for an arbitrary integer k , show that $P(k+1)$ must be true.

Example: Climbing an infinite ladder:

Basic Step: By (1), we can reach rung 1.

Inductive Step: Assume that we can reach rung k .

Then by (2), we can reach the rung $k+1$.

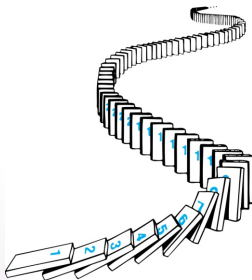
Hence, $P(k) \rightarrow P(k+1)$ is true for all positive integers k .

We can reach every rung of the ladder.

How Mathematical Induction Works

Consider an infinite sequence of dominoes, labeled $1, 2, 3, \dots$, where each domino is standing.

Let $P(n)$ be the proposition that the n -th domino is knocked over.



We know that the first domino is knocked down, i.e., $P(1)$ is true.

We also know that if whenever the k -th domino is knocked over, it knocks over the $(k + 1)$ -st domino, i.e., $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

Hence, all the dominoes are knocked over.

- $P(n)$ is true for all positive integers n .

Proving a Summation Formula

Example: Show that: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

Solution:

BASIS STEP: $P(1)$ is true since $1(1+1)/2 = 1$.

INDUCTIVE STEP: Assume that the inductive hypothesis holds $P(k)$ and then show that $P(k+1)$ holds well.

The inductive hypothesis is $\sum_{i=1}^k i = \frac{k(k+1)}{2}$

Under this assumption,

$$\begin{aligned} P(k+1) &= \sum_{i=1}^{k+1} i = 1 + 2 + \dots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2} \end{aligned}$$

Proving a Summation Formula

Example: Conjecture and prove correct a formula for the sum of the first n positive odd integers. Then prove your conjecture.

Solution: $1 = 1$,

$$1 + 3 = 4, 1 + 3 + 5 = 9, 1 + 3 + 5 + 7 = 16, 1 + 3 + 5 + 7 + 9 = 25$$

We can conjecture that the sum of the first n positive odd integers is n^2 , $1 + 3 + 5 + \dots + (2n - 1) = n^2$

We prove that the conjecture is correct with mathematical induction.

BASIS STEP: $P(1)$ is true since $1^2 = 1$.

INDUCTIVE STEP: $P(k) \rightarrow P(k + 1)$ for every positive integer k .

Induction hypothesis: $1 + 3 + 5 + \dots + (2k - 1) = k^2$

$$1 + 3 + 5 + \dots + (2k - 1) + (2k + 1) = [1 + 3 + 5 + \dots + (2k - 1)] + (2k + 1) = k^2 + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2$$

$P(k + 1)$ follows from $P(k)$.

Important Points

- MI as rule of inference: $(P(1) \wedge \forall k(P(k) \rightarrow P(k+1))) \rightarrow \forall n P(n)$

Domain: set of positive integers.

- We do not assume that $P(k)$ is true for all positive integers.

If we assume that $P(k)$ is true, then $P(k+1)$ must also be true.

- Not always start with the integer 1.

The base step may begin at a starting point b where b is an integer.

Validity of Mathematical Induction

$$(P(1) \wedge \forall k(P(k) \rightarrow P(k+1))) \rightarrow \forall n P(n),$$

Well-ordering property: Every nonempty subset of the set of positive integers has a least element

Validation of mathematical induction:

- $P(1)$ holds, $P(k) \rightarrow P(k+1)$ is true for all positive integers k .
- Assume that $P(n)$ is false for some positive integer.
- S : set of positive integers for which $P(n)$ is false. $S \neq \emptyset$
- By the well-ordering property, S has a least element, say m .
- We know that m cannot be 1 since $P(1)$ holds.
- $m > 1$ and positive, $m - 1$ must be a positive integer.
- Since $m - 1 < m$, it is not in S , so $P(m - 1)$ must be true.
- Since conditional $P(k) \rightarrow P(k + 1)$ for every positive integer k holds, $P(m)$ must also be true, contradicting $P(m)$ being false.
- Hence, $P(n)$ must be true for every positive integer n .

Proving Inequalities

Example: Use mathematical induction to prove that $n < 2^n$ for all positive integers n .

Solution: Let $P(n)$ be the proposition that $n < 2^n$.

BASIS STEP: $P(1)$ is true since $1 < 2^1 = 2$.

INDUCTIVE STEP: Assume $P(k)$ holds, i.e., $k < 2^k$, for an arbitrary positive integer k .

Must show that $P(k+1)$ holds. Since by the inductive hypothesis, $k < 2^k$, it follows that:

$$k + 1 < 2^k + 1 \leq 2^k + 2^k = 2^k(1 + 1) = 2 \cdot 2^k = 2^{k+1}$$

Therefore, $n < 2^n$ holds for all positive integers n .

Proving Inequalities

Example: Use mathematical induction to prove that $2^n < n!$, for every integer $n \geq 4$.

Solution: Let $P(n)$ be the proposition that $2^n < n!$.

BASIS STEP: $P(4)$ is true since $2^4 = 16 < 4! = 24$.

INDUCTIVE STEP: Assume $P(k)$ holds, i.e., $2^k < k!$ for an arbitrary integer $k \geq 4$. To show that $P(k+1)$ holds:

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k < 2 \cdot k! \quad (\text{by the inductive hypothesis}) \\ &< (k+1)k! \quad (\text{since } 2 < (k+1)) \\ &= (k+1)! \end{aligned}$$

Therefore, $2n < n!$ holds for every integer $n \geq 4$.

Note that here the basis step is $P(4)$, since $P(0)$, $P(1)$, $P(2)$, and $P(3)$ are all false.

Proving Divisibility Results

Example: Use mathematical induction to prove that $n^3 - n$ is divisible by 3, for every positive integer n .

Solution: Let $P(n)$ be the proposition that $n^3 - n$ is divisible by 3.

BASIS STEP: $P(1)$ is true since $1^3 - 1 = 0$, which is divisible by 3.

INDUCTIVE STEP: Assume $P(k)$ holds, i.e., $k^3 - k$ is divisible by 3, for an arbitrary positive integer k . To show that $P(k + 1)$ follows:

$$(k + 1)^3 - (k + 1) = (k^3 + 3k^2 + 3k + 1) - (k + 1) = (k^3 - k) + 3(k^2 + k)$$

By the inductive hypothesis, the first term $(k^3 - k)$ is divisible by 3 and the second term is divisible by 3 since it is an integer multiplied by 3. So by part (i) of Theorem 1 in Section 4.1, $(k + 1)^3 - (k + 1)$ is divisible by 3.

Therefore, $n^3 - n$ is divisible by 3, for every positive integer n .

Number of Subsets of a Finite Set

Example: Use mathematical induction to show that if S is a finite set with n elements, where n is a nonnegative integer, then S has 2^n subsets.

Solution: $P(n)$: a set with n elements has 2^n subsets.

Basis Step: $P(0)$ is true, because the empty set has only itself as a subset and $2^0 = 1$.

Inductive Step: Assume $P(k)$ is true for an arbitrary nonnegative integer k .

Inductive Hypothesis: For an arbitrary nonnegative integer k , every set with k elements has 2^k subsets.

Let T be a set with $k + 1$ elements. Then $T = S \cup \{a\}$, where $a \in T$ and $S = T \setminus \{a\}$. Hence $|S| = k$.

For each subset X of S , there are exactly two subsets of T , i.e., X and $X \cup \{a\}$.

By the inductive hypothesis S has 2^k subsets. Since there are two subsets of T for each subset of S , the number of subsets of T is $2 \cdot 2^k = 2^{k+1}$.

Tiling Checkerboards

Example: Show that every $2^n \times 2^n$ checkerboard with one square removed can be tiled using right triominoes.

A right triomino is an L-shaped tile that covers three squares at a time.



Solution: $P(n)$: every $2^n \times 2^n$ checkerboard with one square removed can be tiled using right triominoes.

BS: $P(1)$ is true, because each of the four 2×2 checkerboards with one square removed can be tiled using one right triomino.



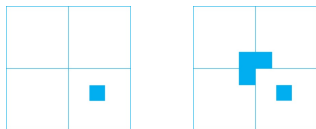
IS: Assume that $P(k)$ is true for every $2^k \times 2^k$ checkerboard, for some positive integer k .

Tiling Checkerboards

Inductive Hypothesis: Every $2^k \times 2^k$ checkerboard, for some positive integer k , with one square removed, can be tiled using right triominoes.

Consider a $2^{k+1} \times 2^{k+1}$ checkerboard with one square removed.

Split this checkerboard into four checkerboards of size $2^k \times 2^k$, by dividing it in half in both directions.



Remove a square from one of the four $2^k \times 2^k$ checkerboards.

By the inductive hypothesis, this board can be tiled.

Also, by the inductive hypothesis, the other three boards can be tiled with the square from the corner of the center of the original board removed. We can then cover the three adjacent squares with a triominoe.

Hence, the entire $2^k + 1 \times 2^k + 1$ checkerboard with one square removed can be tiled using the right triominoes.

An Incorrect “Proof” by Induction

Example: Let $P(n)$ be the statement that every set of n lines in the plane, no two of which are parallel, meet in a common point.

A proof idea:

BASIS STEP: The statement $P(2)$ is true because any two lines in the plane that are not parallel meet at a common point.

INDUCTIVE STEP: The inductive hypothesis is the statement that $P(k)$ is true for the positive integer $k \geq 2$, i.e., every set of k lines in the plane, not two of which are parallel, meet at a common point.

We must show that if $P(k)$ holds, then $P(k+1)$ holds, i.e., if every set of k lines in the plane, no two of which are parallel, $k \geq 2$, meet in a common point, then every set of $k+1$ lines in the plane, no two of which are parallel, meet in a common point.

An Incorrect “Proof” by Induction

Inductive hypothesis: Every set of k lines in the plane, where $k \geq 2$, not two of which are parallel, meet at a common point.

Consider a set of $k + 1$ distinct lines in the plane, no two parallel.

By the inductive hypothesis, the first k of these lines must meet at a common point p_1 and last k of these lines meet at a common point p_2 .

If p_1 and p_2 are different points, all lines containing both of them must be the same line since two points determine a line.

This contradicts the assumption that the lines are distinct. Hence, $p_1 = p_2$ lies on all $k + 1$ distinct lines, and therefore $P(k + 1)$ holds. Assuming that $k \geq 2$, distinct lines meet in a common point, then every $k + 1$ lines meet in a common point.

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Question: There must be an error in this proof since the conclusion is absurd. But where is the error?

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This contradicts the assumption that the lines are distinct. Hence, $p_1 = p_2$ lies on all $k + 1$ distinct lines, and therefore $P(k + 1)$ holds. Assuming that $k \geq 2$, distinct lines meet in a common point, then every $k + 1$ lines meet in a common point.

Question: There must be an error in this proof since the conclusion is absurd. But where is the error?

Answer: $P(k) \rightarrow P(k + 1)$ only holds for $k \geq 3$. It is not the case that $P(2)$ implies $P(3)$. The first two lines must meet at a common point p_1 and the second two must meet at a common point p_2 . They do not have to be the same point since only the second line is common to both sets of lines.

Template for Proofs by Induction

- 1 Express the statement that is to be proved in the form “for all $n \geq b, P(n)$ ” for a fixed integer b .
- 2 Write the words “Basis Step.” Then show that $P(b)$ is true, taking care that the correct value of b is used.
- 3 Write the words “Inductive Step.”
- 4 State and clearly identify the inductive hypothesis in the form “assume that $P(k)$ is true for an arbitrary fixed integer $k \geq b$.”
- 5 State what needs to be proved under the assumption that the inductive hypothesis is true. That is, write out what $P(k+1)$ says.
- 6 Prove the statement $P(k+1)$ making use the assumption $P(k)$. Be sure that your proof is valid for all integers k with $k \geq b$, taking care that the proof works for small values of k , including $k = b$.
- 7 Clearly identify the conclusion of the inductive step, such as by saying “this completes the inductive step.”
- 8 After completing the basis step and the inductive step, state the conclusion, namely that by mathematical induction, $P(n)$ is true for all integers n with $n \geq b$.

Strong Induction

Strong Induction: To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, complete two steps:

Basis Step: Verify that the proposition $P(1)$ is true.

Inductive Step: Show the conditional statement $[P(1) \wedge P(2) \wedge \dots \wedge P(k)] \rightarrow P(k+1)$ holds for all positive integers k .

Strong Induction is sometimes called the second principle of mathematical induction or complete induction.

Strong Induction: the Infinite Ladder

Strong induction tells us that we can reach all rungs if:

we can reach the first rung of the ladder.

For every integer k , if we can reach the first k rungs, then we can reach the $(k + 1)$ -st rung.

To conclude that we can reach every rung by strong induction:

BASIS STEP: $P(1)$ holds

INDUCTIVE STEP: Assume $P(1) \wedge P(2) \wedge \dots \wedge P(k)$ holds for an arbitrary integer k , and show that $P(k + 1)$ must also hold.

We will have then shown by strong induction that for every positive integer n , $P(n)$ holds, i.e., we can reach the n -th rung of the ladder.

Proof using Strong Induction

Example: Suppose we can reach the first and second rungs of an infinite ladder, and we know that if we can reach a rung, then we can reach two rungs higher. Prove that we can reach every rung. (Try this with mathematical induction.)

Solution: Prove the result using strong induction.

BASIS STEP: We can reach the first step.

INDUCTIVE STEP: The inductive hypothesis is that we can reach the first k rungs, for any $k \geq 2$. We can reach the $(k + 1)$ -st rung since we can reach the $(k - 1)$ st rung by the inductive hypothesis.

Hence, we can reach all rungs of the ladder.

Which Form of Induction Should Be Used?

We can always use strong induction instead of mathematical induction.

But there is no reason to use it if it is simpler to use mathematical induction.

The principles of mathematical induction, strong induction, and the well-ordering property are equivalent.

Sometimes, it is clear how to proceed using one of the three methods, but not the other two.

Proof using Strong Induction

Example: Prove that every amount of postage of 12 cents or more can be formed using just 4-cent and 5-cent stamps.

Solution: Let $P(n)$ be the proposition that postage of n cents can be formed using 4-cent and 5-cent stamps.

BASIS STEP: $P(12)$, $P(13)$, $P(14)$, and $P(15)$ hold.

$P(12)$ uses three 4-cent stamps.

$P(13)$ uses two 4-cent stamps and one 5-cent stamp.

$P(14)$ uses one 4-cent stamp and two 5-cent stamps.

$P(15)$ uses three 5-cent stamps.

INDUCTIVE STEP: The inductive hypothesis states that $P(j)$ holds for $12 \leq j \leq k$, where $k \geq 15$. Assuming the inductive hypothesis, it can be shown that $P(k+1)$ holds. Using the inductive hypothesis, $P(k-3)$ holds since $k-3 \geq 12$. To form postage of $k+1$ cents, add a 4-cent stamp to the postage for $k-3$ cents. Hence, $P(n)$ holds for all $n \geq 12$.

Same example using induction

Example: Prove that every amount of postage of 12 cents or more can be formed using just 4-cent and 5-cent stamps.

Solution: Let $P(n)$ be the proposition that postage of n cents can be formed using 4-cent and 5-cent stamps.

BASIS STEP: Postage of 12 cents can be formed using three 4-cents.

INDUCTIVE STEP: The inductive hypothesis $P(k)$ for any positive integer k is that postage of k cents can be formed using 4-cent and 5-cent. To show $P(k + 1)$ where $k \geq 12$, we consider two cases:

If at least one 4-cent has been used, then a 4-cent can be replaced with a 5-cent to yield a total of $k + 1$ cents.

Otherwise, no 4-cent have been used and at least three 5-cent were used. Three 5-cent can be replaced by four 4-cent to yield a total of $k + 1$ cents.

Hence, $P(n)$ holds for all $n \geq 12$.

Fundamental Theorem of Arithmetic

Theorem: If n is an integer greater than 1, then n can be written as the product of primes.

Proof: $P(n)$: n can be written as a product of primes.

BASIS STEP: $P(2)$ is true since 2 itself is prime.

INDUCTIVE STEP: The inductive hypothesis is that $P(j)$ is true for all integers j with $2 \leq j \leq k$. To show that $P(k+1)$ must be true under this assumption, two cases must be considered.

If $k+1$ is prime, then $P(k+1)$ is true.

Otherwise, $k+1$ is composite and can be written as the product of two positive integers a and b with $2 \leq a \leq b < k+1$. By the inductive hypothesis, a and b can be written as the product of primes and, therefore, $k+1$ can also be written as the product of those primes. Hence, it has been shown that every integer greater than 1 can be written as the product of primes.

Well-Ordering Property

Well-ordering property: Every nonempty set of nonnegative integers has a least element.

The well-ordering property is an axiom of the positive integers.

The well-ordering property can be used directly in proofs.

The well-ordering property can be generalized.

Definition: A set is well ordered if every subset has a least element.

$\mathbb{N}$ is well ordered under $\leq$.

The set of finite strings over an alphabet using lexicographic ordering is well ordered.

We will see a generalization of induction to sets other than the integers in the next section.

Well-Ordering Property

Example: Use the well-ordering property to prove the division algorithm, which states that if a is an integer and d is a positive integer, then there are unique integers q and r with $0 \leq r < d$, such that $a = dq + r$.

Solution: Let S be the set of non-negative integers of the form $a - dq$, where q is an integer.

The set is nonempty since $-dq$ can be made as large as needed. Take q large negative.

By the well-ordering property, S has the least element $r = a - dq_0$. The integer r is non-negative. It must also be the case that $r < d$. If it were not, then there would be a smaller non-negative element in S , that is, $a - d(q_0 + 1) = a - dq_0 - d = r - d > 0$.

Therefore, there are integers q and r with $0 \leq r < d$.

Recursive definition

Recursive or inductive definition: A recursive or inductive definition of a function consists of two steps.

Base step: Specify the value of the function at zero.

Recursive step: Give a rule for finding its value at an integer from its values at smaller integers.

A function $f : \mathbb{N} \rightarrow \mathbb{N}$ corresponds to sequence $a_0, a_1, \dots$ where $a_i = f(i)$.

Example: Give a recursive definition of the factorial function $n!$.

Solution: $f(0) = 1, f(n) = n * f(n - 1)$

Example: Give a recursive definition of the Fibonacci numbers.

Solution: $f(0) = 0, f(1) = 1, f(n) = f(n - 2) + f(n - 1)$

Recursive definition

Example: Suppose that f is defined recursively by $f(0) = 3$,
 $f(n+1) = 2f(n) + 3$.

Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$.

Solution: From the recursive definition it follows that

$$f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9,$$

$$f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21,$$

$$f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45,$$

$$f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93.$$

Recursive definition

Example: Give a recursive definition of a^n , where a is a nonzero real number and n is a nonnegative integer.

Solution: Basis Step: $a_0 = 1$.

Recursive Step: The rule for finding a^{n+1} from a^n , namely, $a^{n+1} = a \cdot a^n$, for $n = 0, 1, 2, 3, \dots$, is given.

Example: Give a recursive definition of $\sum_{k=0}^n a_k$

Solution: The first part of the recursive definition is $\sum_{k=0}^0 a_k = a_0$ The second part is $\sum_{k=0}^{n+1} a_k = \sum_{k=0}^n a_k + a_{n+1}$

Recursively defined sets and structures

H.W.